

15. Body Temperature and Heart Rate Is there any relationship between these two variables? To find out, we randomly selected 12 people from a data set constructed by Allen Shoemaker (*Journal of Statistics Education*) and recorded their body temperature and heart rate.¹⁹

Person	1	2	3	4	5	6
Temperature (degrees)	96.3	97.4	98.9	99.0	99.0	96.8
Heart Rate (beats per minute)	70	68	80	75	79	75

Person	7	8	9	10	11	12
Temperature (degrees)	98.4	98.4	98.8	98.8	99.2	99.3
Heart Rate (beats per minute)	74	84	73	84	66	68

- Find the correlation coefficient r , relating body temperature to heart rate.
- Is there sufficient evidence to indicate that there is a correlation between these two variables? Test at the 5% level of significance.

a.

$$r = 0.1741$$

b.

$$H_0: \rho = 0, H_a: \rho > 0$$

$$\alpha = 0.05$$

$$t_{STAT} = r \cdot \sqrt{\frac{n-2}{1-r^2}} \sim t_{n-2, 0.05}$$

$$t^* = 0.1741 \cdot \sqrt{\frac{10}{1-0.1741^2}} = 0.3591$$

$$RR: \{t \mid |t| > t_{10, 0.05}\} = \{t \mid |t| > 2.2281\}$$

$$0.3591 < 2.2281 \Rightarrow \text{non-Reject } H_0$$

$\Rightarrow$ There isn't sufficient evidence to indicate that there's a correlation between x, y .